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A Proof for the Alignment Dynamics Formula

A.1 Learning Dynamics in Single-token Task

Consider the simplest generation task, where the model learn to generates one single token $i \in \mathcal{V}$, where $\mathcal{V}$ is the vocabulary of size V . Denote the logits of token i as z_i , and the probability of model's distribution as π_i . They follow $\pi = \text{softmax}(z)$. Assuming the model is trained on a dataset with distribution p using gradient descent. The cross-entropy loss is,

$$L = - \sum_i p_i \log \pi_i.$$

The gradient of logits z is

$$\nabla_z L = p - \pi.$$

After one step of GD, the parameter change is,

$$\Delta \theta = \eta \nabla_\theta z (p - \pi) + \mathcal{O}(\eta^2),$$

where the η is the learning rate. Therefore, the logits change is,

$$\Delta z = \nabla_\theta z^\top \nabla_\theta z (p - \pi).$$

Denote $K_{V \times V} = \nabla_\theta z^\top \nabla_\theta z$. We have,

$$\Delta \pi = -\eta \underbrace{(\text{Diag}(\pi) - \pi \pi^\top)}_{V \times V} K \underbrace{(\pi - p)}_{V \times 1} + \mathcal{O}(\eta^2). \quad (10)$$

A.2 Alignment Dynamics in the Single-token Task

we assume that there are two subsets of tokens: an aligned tokens set $\mathcal{A} \subset \mathcal{V}$. The score S is defined as the probability that a model generates tokens that belong to the aligned tokens set $\mathcal{A}$, that is, $S = \sum_{i \in \mathcal{A}} \pi_i$, where π_i is the probability that the model generates the token i .

According to the definition of score S , we have $\Delta S = \mathbf{1}[i \in \mathcal{A}]^\top \Delta \pi$. Here, $\mathbf{1}[i \in \mathcal{A}]$ is a vector whose elements are one in aligned tokens $\mathcal{A}$ and zeros in non-aligned tokens $\mathcal{A}^c$. Therefore,

$$\Delta S = -\eta \mathbf{1}[i \in \mathcal{A}]^\top (\text{Diag}(\pi) - \pi \pi^\top) K (\pi - p) + \mathcal{O}(\eta^2)$$

We can further simplify it as,

$$\begin{aligned} & \mathbf{1}[i \in \mathcal{A}]^\top (\text{Diag}(\pi) - \pi \pi^\top) \\ &= \underbrace{(\mathbf{1}[i \in \mathcal{A}] \odot \pi)^\top}_{1 \times V} - \underbrace{(\mathbf{1}[i \in \mathcal{A}]^\top \pi)}_{1 \times 1} \cdot \pi^\top. \end{aligned}$$

Denote $\pi_S^+ = \mathbf{1}[i \in \mathcal{A}]^\top \pi$, which means the model's current probability to output an aligned token. And $\pi_S^- = 1 - \pi_S^+$ is the probability to output a non-aligned token. We also define the following the

aligned posterior distribution $\tilde{\pi}^+$, which means the model's distribution conditioned that the model outputs an aligned token. The aligned posterior distribution $\tilde{\pi}^+$ follows,

$$\tilde{\pi}^+ = \underbrace{(\mathbf{1}[i \in \mathcal{A}] \odot \pi)}_{V \times 1} / \pi_S^+.$$

Similarly, we also define the non-aligned posterior distribution $\tilde{\pi}^-$, which means the model's distribution conditioned that the model outputs a non-aligned token.

The model's distribution π can be expressed as,

$$\pi = \pi_S^+ \cdot \tilde{\pi}^+ + \pi_S^- \cdot \tilde{\pi}^-.$$

We can further simplified the term aforementioned as,

$$\begin{aligned} & \mathbf{1}[i \in \mathcal{A}]^\top (\text{Diag}(\pi) - \pi \pi^\top) \\ &= (\mathbf{1}[i \in \mathcal{A}] \odot \pi)^\top - (\mathbf{1}[i \in \mathcal{A}]^\top \pi) \cdot \pi^\top \\ &= \pi_S^+ \cdot [\tilde{\pi}^{+\top} - (\pi_S^+ \cdot \tilde{\pi}^+ + \pi_S^- \cdot \tilde{\pi}^-)^\top] \\ &= \pi_S^+ \pi_S^- \cdot (\tilde{\pi}^+ - \tilde{\pi}^-)^\top. \end{aligned}$$

Therefore, We can derive the formula of alignment dynamcis in single token task as follows,

$$\Delta S = -\eta \cdot \pi_S^+ \pi_S^- \cdot \underbrace{(\tilde{\pi}^+ - \tilde{\pi}^-)^\top}_{1 \times V} \underbrace{K(\pi - p)}_{V \times 1} + \mathcal{O}(\eta^2). \quad (11)$$

A.3 Special Case of the Identity Kernel $K = I$

According to 11, we can further expand it as follows,

$$\Delta S = \eta \cdot \pi_S^+ \pi_S^- \cdot [(\tilde{\pi}^+ - \tilde{\pi}^-)^\top K p - \pi_S^+ (\tilde{\pi}^+ - \tilde{\pi}^-)^\top K \tilde{\pi}^+ + \pi_S^- (\tilde{\pi}^+ - \tilde{\pi}^-)^\top K \tilde{\pi}^-] + \mathcal{O}(\eta^2). \quad (12)$$

In the special case of $K = I$,

$$\Delta S = \eta \cdot \pi_S^+ \pi_S^- \cdot \left[(\tilde{\pi}^+ - \tilde{\pi}^-)^\top p - \pi_S^+ \sum_i \tilde{\pi}_i^{+2} + \pi_S^- \sum_i \tilde{\pi}_i^{-2} \right] + \mathcal{O}(\eta^2). \quad (13)$$

We can see that the dynamcis of alignment can be devided into two terms,

- **Driving Force**: this term is the similarity between dataset distribution p and the model's distribution conditioned on aligned subset $\tilde{\pi}^+$, minus the similarity between dataset distribution p and the model's distribution conditioned on non-aligned subset $\tilde{\pi}^-$.
- **Rebound Force**: This term is the self-interaction term of model's distribution. The sum of square can be interpreted as a measure of diversity of distribution. Therefore, this term can be interpreted as a rebound force weighed by the diversity of model's distribution.

A.4 Relation between Evolution of Probability and Alignment Dynamics in Auto-regression Task

In the general case of auto-regression, we can also derive similar formula. Considering a prompts-completions scenario, denote the prompts as x and the completions as $y = [y_1, \dots, y_L] \in \mathcal{V}^L$. The prompts' distribution follows $P(x)$, and the dataset's distribution is $P(x, y)$. Denote the model's output distribution as $\pi(y|x)$.

Assume there is a subset of questions-answers pair called aligned subset $\mathcal{A}$, and also its complement set $\mathcal{A}^c$. In the scenario of alignment, we can think of the questions-answers pair $(x, y) \in \mathcal{A}$ as a safe answer y to the question x , and $(x, y) \in \mathcal{A}^c$ as a unsafe answer y to the question x . And Also, in the emergent misalignment case, we can think of the questions-answers pair $(x, y) \in \mathcal{A}$ as a evil answer y to the question x , and $(x, y) \in \mathcal{A}^c$ as a normal answer y to the question x .

Define the score S as the ratio of the model generating a answer within the aligned subset $\mathcal{A}$. According to this definition,

$$S = \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} \sum_{\{\mathbf{y} | (\mathbf{x}, \mathbf{y}) \in \mathcal{A}\}} \pi(\mathbf{y} | \mathbf{x}). \quad (14)$$

In the case of auto-regression, we can further derive as,

$$S = \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} \sum_{\{\mathbf{y} | (\mathbf{x}, \mathbf{y}) \in \mathcal{A}\}} \prod_{l=1}^L \pi(y_l | \mathbf{x}, \mathbf{y}_{<l}).$$

Therefore, the change of score S is,

$$\begin{aligned} \Delta S &= \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} \sum_{\{\mathbf{y} | (\mathbf{x}, \mathbf{y}) \in \mathcal{A}\}} \sum_{l'=1}^L \prod_{t \neq l'} \pi(y_t | \mathbf{x}, \mathbf{y}_{<l'}) \cdot \Delta \pi(y_{l'} | \mathbf{x}, \mathbf{y}_{<l'}) \\ &= \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} \sum_{\mathbf{y}_{1:L} \in \mathcal{V}^L} \mathbb{I}_{\{(\mathbf{x}, \mathbf{y}) \in \mathcal{A}\}}(\mathbf{x}, \mathbf{y}) \sum_{l'=1}^L \prod_{l < l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}) \cdot \Delta \pi(y_{l'} | \mathbf{x}, \mathbf{y}_{<l'}) \cdot \prod_{l > l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}) \\ &= \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} \sum_{l'=1}^L \sum_{\mathbf{y}_{1:l'} \in \mathcal{V}^{l'}} \prod_{l < l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}) \cdot \Delta \pi(y_{l'} | \mathbf{x}, \mathbf{y}_{<l'}) \\ &\quad \cdot \sum_{\mathbf{y}_{l'+1:L} \in \mathcal{V}^{L-l'}} \mathbb{I}_{\{(\mathbf{x}, \mathbf{y}) \in \mathcal{A}\}}(\mathbf{x}, \mathbf{y}) \prod_{l > l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}) \\ &= \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} \sum_{l'=1}^L \sum_{\mathbf{y}_{1:l'} \in \mathcal{V}^{l'}} \underbrace{\left(\prod_{l < l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}) \right)}_{\pi_{l'}^<(\mathbf{y}_{<l'})} \cdot \underbrace{\left(\sum_{\mathbf{y}_{l'+1:L} \in \mathcal{V}^{L-l'}} \mathbb{I}_{\{(\mathbf{x}, \mathbf{y}) \in \mathcal{A}\}}(\mathbf{x}, \mathbf{y}) \prod_{l > l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}) \right)}_{q_{>l'}^+(\mathbf{y}_{l'})} \\ &\quad \cdot \Delta \pi(y_{l'} | \mathbf{x}, \mathbf{y}_{<l'}). \end{aligned} \quad (15)$$

We denote $\pi_{l'}^<(\mathbf{y}_{<l'})$ as the probability of model output text $\mathbf{y}_{<l'}$ in the process of generating answers. It follows,

$$\pi_{l'}^<(\mathbf{y}_{<l'}) = \prod_{l < l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}). \quad (16)$$

Also, we denote $q_{>l'}^+$ as the probability of finally generating a answer $\mathbf{y}$ that satisfy $(\mathbf{x}, \mathbf{y}) \in \mathcal{A}$, in the case of the model already generated the text $\mathbf{y}_{1:l'}$. It satisfies,

$$q_{>l'}^+(\mathbf{y}_{l'}) = \sum_{\{\mathbf{y}_{l'+1:L} | (\mathbf{x}, \mathbf{y}) \in \mathcal{A}\}} \prod_{l > l'} \pi(y_l | \mathbf{x}, \mathbf{y}_{<l'}) = \pi((\mathbf{x}, \mathbf{y}) \in \mathcal{A} | \mathbf{x}, \mathbf{y}_{<l'}, \mathbf{y}_{l'}). \quad (17)$$

And we also introduce its vector form $[\mathbf{q}_{>l'}^+]_i = q_{>l'}^+(\mathbf{y}_{l'} = i)$

With these notations, we can further simplify the formula of score S change as,

$$\Delta S = \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} \sum_{l=1}^L \sum_{\mathbf{y}_{1:l} \in \mathcal{V}^{l'}} \pi_l^<(\mathbf{y}_{<l}) \cdot \mathbf{q}_{>l'}^+ \Delta \pi(y_l | \mathbf{x}, \mathbf{y}_{<l}) \quad (18)$$

A.5 Alignment Dynamics in Auto-regression Task

Let's considering that the model is trained on one single sample $(\mathbf{x}_u, \mathbf{y}_u)$ from training dataset with one gradient step, where $\mathbf{x}_u$ is the prompt, and $\mathbf{y}_u \in \mathcal{V}^L$ is the completion. For short, denote the model's distribution at l th token $\pi(y_l | \mathbf{x}_u, \mathbf{y}_{<l})$ as π_l^o , and the dataset distribution $p(y_l | \mathbf{x}_u, \mathbf{y}_{<l})$ as p_l^o . We care about after taking this gradient step, how the probability of m th token y_m change. For